

BEE ACADEMY — SAMPLE QUESTION BANK

23 Questions | 5 Subjects | Multiple Choice & True/False

[MATHEMATICS]

1. A quadratic equation $ax^2 + bx + c = 0$ has at most how many real solutions?

- A. One solution
- B. Two solutions
- C. Three solutions
- D. No solution

Answer: B | A quadratic equation has at most 2 real roots (when discriminant ≥ 0).

2. What is the value of $\sin(90 \text{ degrees})$?

- A. 0
- B. 1
- C. -1
- D. $\sqrt{2}/2$

Answer: B | $\sin(90 \text{ degrees}) = 1$ is the maximum value of the sine function.

3. The area of a circle with radius r is calculated by which formula?

- A. $2\pi r$
- B. πr^2
- C. $4\pi r^2$
- D. πd

Answer: B | $\text{Area} = \pi r^2$ is the formula for the area of a circle.

4. The sum of all angles in a triangle equals how many degrees?

- A. 90 degrees
- B. 270 degrees
- C. 180 degrees
- D. 360 degrees

Answer: C | The sum of three interior angles of any triangle is always 180 degrees.

5. What is the derivative of $f(x) = x^3$?

- A. $3x$
- B. x^2
- C. $3x^2$
- D. $2x^3$

Answer: C | By the power rule: $f'(x) = 3x^2$.

[PHYSICS]

6. TRUE or FALSE: Newton's First Law states that an object at rest remains at rest if no net force acts on it.

- A. True
- B. False

Answer: A | True - this is the law of inertia.

7. TRUE or FALSE: The formula for velocity is $v = s \times t$ (distance multiplied by time).

- A. True
- B. False

Answer: B | False - the correct formula is $v = s/t$ (distance divided by time).

8. What is the SI unit of force?

- A. Kilogram (kg)
- B. Newton (N)
- C. Joule (J)
- D. Pascal (Pa)

Answer: B | Force is measured in Newtons (N), named after Isaac Newton.

9. What is the approximate speed of light in a vacuum?

- A. 300,000 km/h
- B. 300,000 km/s
- C. 3,000,000 km/s

D. 30,000 km/s

Answer: B | The speed of light c is approximately $3 \times 10^8 \text{ m/s} = 300,000 \text{ km/s}$.

10. What is the formula for kinetic energy of an object?

- A. $KE = mv$
- B. $KE = mv^2$
- C. $KE = (1/2)mv^2$
- D. $KE = 2mv^2$

Answer: C | Kinetic energy $KE = (1/2)mv^2$, where m is mass and v is velocity.

[CHEMISTRY]

11. What element has the chemical symbol Fe?

- A. Fluoride
- B. Fluorine
- C. Iron
- D. Phosphorus

Answer: C | Fe stands for Iron (from the Latin word Ferrum).

12. What is the chemical formula for water?

- A. H_2O_2
- B. HO
- C. H_2O
- D. H_3O

Answer: C | Water consists of 2 hydrogen atoms and 1 oxygen atom: H_2O .

13. What is the atomic number of Oxygen?

- A. 6
- B. 7
- C. 8
- D. 9

Answer: C | Oxygen (O) has atomic number $Z = 8$.

14. Which metal has the highest electrical conductivity?

- A. Gold (Au)
- B. Copper (Cu)
- C. Silver (Ag)
- D. Aluminum (Al)

Answer: C | Silver has the highest electrical conductivity of all metals.

15. TRUE or FALSE: The periodic table was created by Mendeleev.

- A. True
- B. False

Answer: A | True - Dmitri Mendeleev published the first periodic table in 1869.

[ENGLISH]

16. Which of the following words is an adjective?

- A. Run
- B. Beautiful
- C. Quickly
- D. Happiness

Answer: B | "Beautiful" is an adjective that describes a quality or characteristic.

17. TRUE or FALSE: The Present Perfect tense is used for actions that happened in the past and still have relevance to the present.

- A. True
- B. False

Answer: A | True - Present Perfect connects a past action to the present moment.

18. Complete the sentence: "This house _____ in 1990." (passive voice of: They built this house in 1990)

- A. was build
- B. were built
- C. was built
- D. is built

Answer: C | Past Simple passive voice: was/were + past participle. Singular subject uses "was built".

19. What is the correct past tense of "go"?

- A. goed
- B. gone
- C. went
- D. go

Answer: C | "Went" is the simple past tense of the irregular verb "go".

[HISTORY & GEOGRAPHY]

20. What is the capital city of Vietnam?

- A. Ho Chi Minh City
- B. Da Nang
- C. Hanoi
- D. Hue

Answer: C | Hanoi is the capital of the Socialist Republic of Vietnam.

21. On which date did Vietnam declare its independence?

- A. April 30, 1975
- B. September 2, 1945
- C. August 19, 1945
- D. May 7, 1954

Answer: B | On September 2, 1945, President Ho Chi Minh read the Declaration of Independence at Ba Dinh Square.

22. Which is the longest river in the world?

- A. Amazon
- B. Nile
- C. Yangtze
- D. Mississippi

Answer: B | The Nile River in Africa is approximately 6,650 km long, the longest in the world.

23. Which continent has the largest land area?

- A. Americas
- B. Africa
- C. Europe
- D. Asia

Answer: D | Asia covers approximately 44.6 million km², making it the largest continent.